MD. JAMIL KHAN

Full Stack AI and Automation Engineer

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PROFESSIONAL SUMMARY

Full Stack AI Engineer, 1+ year. LLM, RAG, AI agents, workflow automation. FastAPI, Next.js, n8n, LangGraph, CrewAI. Docker, CI/CD, VPS, AWS. International client experience. Published ML/DL researcher (Q1, Q2 journals).

CORE SKILLS

- **AI and LLM:** LLM Integration, AI Agents, Multi-Agent Systems, RAG, Prompt Engineering, Tool Calling, MCP, LangChain, LangGraph, CrewAI, OpenAI (GPT-4o), Gemini, Ollama, Hugging Face, Vector Databases (FAISS, Pinecone, pgvector), Embeddings
- **Automation:** n8n, Workflow Automation, OpenClaw, Headless Browser Automation, Web Scraping, API Integration, Webhooks, Email Automation (SMTP), Meta Ads Automation, Meta Graph API
- **Voice and Vision:** ElevenLabs, Vapi AI, Twilio, OCR, EasyOCR, YOLO11, OpenCV, Computer Vision
- **Backend:** Python, FastAPI, Flask, Laravel, PHP, Node.js, REST API, WebSockets, Asyncio, Microservices, Redis, Background Workers, JWT, Stripe
- **Frontend:** Next.js, React, TypeScript, JavaScript, Tailwind CSS
- **Database:** PostgreSQL, MySQL, MongoDB, Supabase, SQLAlchemy, Redis
- **ML and DL:** PyTorch, TensorFlow, Scikit-learn, XGBoost, LightGBM, CNN, Vision Transformers
- **DevOps:** Docker, Docker Compose, GitHub Actions, CI/CD, Nginx, Linux, VPS, AWS EC2

WORK EXPERIENCE

AI Engineer (Full Stack) **July 2026 – Present**
Nibiz soft, Dhaka, Bangladesh (On-site)

- End-to-end ownership, from requirements to VPS deployment: FastAPI, Next.js, Docker, CI/CD
- **LeadGen AI:** automated lead discovery, website audit (135 parameters), email, Meta Ads automation and ElevenLabs voice calls, replacing manual research and outreach
- **AI CCTV toll system:** real-time vehicle detection, number plate OCR and automatic toll calculation (YOLO11, EasyOCR, Flask, MySQL)
- DevOps: Docker Compose, Nginx, GitHub Actions, VPS

AI Engineer **Nov 2025 – Jun 2026**
Softvence, Dhaka, Bangladesh (On-site)

- RAG pipelines, AI agents, multi-agent systems: GPT-4o, LangGraph, CrewAI, FastAPI
- **n8n** workflow automation: booking, support triage, ticket deduplication across 5+ APIs
- Multi-tenant AI agent SaaS: 15 agents, 4 messaging channels
- **BilledSmart:** OCR and RAG bill analysis, 100% of manual bill review eliminated
- **WhatsApp booking bot:** text, audio and image input, 0 manual intervention
- Vector search (FAISS, Pinecone, pgvector), LangSmith monitoring, multi-LLM (OpenAI, Gemini, Groq, Grok)
- Docker, CI/CD, AWS EC2, VPS; international client calls and delivery reporting

Research Assistant **Oct 2024 – May 2025**
Metropolitan University, Sylhet, Bangladesh

- ML/DL: heart disease prediction 96.36%; Swin Transformer + ResNet50 cancer image classification, up to 99.23%

PROJECTS

LeadGen AI: Autonomous B2B Lead Generation and Outreach Platform

- **OpenClaw:** headless Chromium crawler, screenshots, contacts, tech stack, deduplication
- **Lead scoring:** 135 parameters (SEO, speed, mobile, SSL, tracking pixels)
- **Email automation:** Ollama, Gemini, approval queue, SMTP (Gmail, AWS SES, SendGrid, Zoho)
- **Meta Ads automation:** Meta Graph API, ad copy, creative briefs, Meta Webhooks for inbound leads
- **Voice agent:** ElevenLabs, multi-language (Bengali, English, Arabic, Spanish, etc.), objection handling, booking
- **Dashboard:** Next.js 15, WebSockets, real-time worker logs
- **Result:** discovery to outreach runs automatically; a human approves every email before sending
- *Stack:* Python, FastAPI, SQLAlchemy (async), PostgreSQL, Redis, Next.js, React, TypeScript, JWT, Docker Compose, GitHub Actions, N8N

FinSight: Enterprise Banking Intelligence & Analytics Platform

- **Statistical ML Pipelines:** Developed custom Scikit-Learn transformers performing statistical screening (χ^2, correlation), financial domain ratio engineering with $log1p$ dampening, and adaptive 95% PCA reduction.
- **Credit Underwriting & Fraud:** Engineered XGBoost credit risk scoring (ROC-AUC: 0.938) and fraud defense on 0.17% extreme imbalance (PR-AUC: 0.854) capturing $>$85% fraud at $<$0.05% FPR via calibrated threshold routing.
- **Marketing & Customer Analytics:** Built leakage-free prospective deposit scoring cutting dialer costs by $\sim$40%, churn prediction (ROC-AUC: 0.862), 4-cluster RFM segmentation, and 28-day revenue forecasting (MAE: £10,854).
- **Serving & Architecture:** Deployed a 12-page Streamlit analytics portal, low-latency FastAPI REST service, declarative YAML configs, and a complete automated test suite (Pytest, 100% pass).
- *Stack:* Python 3.11, Scikit-Learn, XGBoost, FastAPI, Streamlit, Plotly, Pandas, NumPy, SciPy, Statsmodels, Pydantic, Joblib, Pytest

AI-Powered Multi-Tenant School ERP SaaS

- **Core ERP:** multi-tenant architecture with tenant isolation, RBAC, fee/payroll automation, 25+ PDF reports
- **GenAI engine:** FastAPI microservice with OpenAI and Gemini for automated question generation and RAG chatbot
- **CCTV proctoring:** real-time vision-based exam surveillance detecting suspicious activities and unauthorized behavior
- **Reliability:** decoupled asynchronous setup, core ERP keeps running if AI/vision services go offline
- *Stack:* Laravel, PHP 8.3, MySQL, FastAPI, Python, OpenCV, OpenAI, Gemini, Docker Compose, Tailwind CSS

VisionPark AI: Smart AI Parking and Vehicle Surveillance System

- **Detection:** fine-tuned YOLO11x on 21 custom vehicle classes (CNG, auto-rickshaw, car), 150 epochs, 1280 px, AdamW, mosaic and copy-paste augmentation
- **OCR pipeline:** 2-stage ANPR for Bengali (BRTA standard) and English plates via EasyOCR; multi-stream RTSP, NVR and webcam ingestion
- **Tracking and logic:** automated same-plate entry/exit matching, duration calculation, slot allocation, overstay alerts
- **Reliability:** duplicate-logging prevention via IoU tracking and plate cooldowns; threaded Flask REST APIs with SQLite/MySQL fallback
- **Dashboard:** full-stack web SPA with real-time Chart.js occupancy metrics, movement filter logs, automated PDF/Excel reports
- *Stack:* YOLO11x, Ultralytics, PyTorch, OpenCV, EasyOCR, Python, Flask, SQLite, MySQL, JavaScript, Docker

EDUCATION

M.Sc. in Applied Statistics and Data Science — **2026 – Present**
Jahangirnagar University, Dhaka, Bangladesh

B.Sc. in Computer Science and Engineering — **2021 – 2025**
Metropolitan University, Sylhet, Bangladesh — CGPA: 3.67 / 4.00

PUBLICATIONS

DSVTLA: Swin Vision Transformer Transfer Learning for Multi-Type Cancer Image Classification
Computers in Biology and Medicine (Q1, Elsevier), co-author, under review, 2026

- **Model:** hybrid Swin-Base Transformer + ResNet50, transfer learning, fine-tuning (PyTorch)
- **Data:** 5 public datasets, 58,885 images (breast, oral, lung-colon, kidney CT, leukemia blood smear; original and segmented), 70/15/15 split
- **Benchmark:** 9 baselines (DenseNet121/201, InceptionV3, ResNet50, EfficientNetB3, ViT, Swin), one unified pipeline
- **Explainable AI:** LIME, SHAP, occlusion sensitivity
- **Result:** up to 99.23% accuracy (breast cancer), 100% test accuracy (lung-colon, kidney, segmented leukemia)

Harmonization of Heart Disease Dataset for Accurate Diagnosis
CMC: Computers, Materials & Continua (Q2, SCI, Scopus), published Mar 2025

- **Model:** RF, XGBoost, Gradient Boosting, LightGBM, CatBoost, SVM, MLP, ensembles, 1D CNN (TensorFlow, Scikit-learn)
- **Method:** data fusion from several regions, outlier handling, imputation, feature interaction (correlation, chi-square), PCA (95% variance)
- **Result:** 1D CNN 96.36% accuracy (specificity 99.51%), 7% better than without feature engineering; classical ML 88–92%